CampusEats — Live Catalogue Service

End-to-End Three Layer Web Architecture Implementation

Fully Functional • Verified

📌 Application Overview

CampusEats is a modern, responsive web application serving an active kitchen catalogue. The application implements a decoupled three-layer architecture where the browser interface interacts seamlessly with the backend service to retrieve menu items and register new dishes in real time.

🏗️ Three-Layer Architecture

Layer	Component	Key Responsibilities
Front-End Layer	`public/index.html`, `public/style.css`	Modern responsive UI with food category icons, vegetarian/non-vegetarian indicators, type-ahead live search filter, and instant toast notifications.
Service Layer	`server.js` (Port 3000)	RESTful route handlers parsing JSON payloads, serving static assets, and providing standard HTTP status codes (`200 OK` , `201 Created` , `404 Not Found` , `422 Unprocessable`).
Data Layer	`items` Array	In-memory data store holding active dish records (`id` , `name` , `price`).

🔌 Verified Endpoints

Method	Endpoint	Description	Response Code
GET	`/items`	Retrieves the complete active kitchen menu.	`200 OK`
GET	`/items/:id`	Retrieves a single dish by unique ID; returns 404 if not found.	`200 OK` / `404 Not Found`
POST	`/items`	Creates and adds a new dish; validates required name.	`201 Created` / `422 Unprocessable`

🎬 Deliverables & Working Demonstration

- **Unified MP4 Video:** `demonstration.mp4` — High-definition single walkthrough video (compatible natively with Windows Media Player).
- **Interactive Player:** `demonstration.html` — Modern HTML player with playback controls and live specs.
- **Animated Preview:** `demonstration.gif` — Embedded auto-looping animation demonstrating live browsing, search, and dish additions.
- **Documentation:** `README.md` & `ReadMe.pdf` — Complete architecture, setup instructions, and verified curl test logs.